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Paleomagnetism and $^{40}Ar/^{39}Ar$ ages of mafic dikes from Salvador (Brazil): new constraints on the São Francisco craton APW path between 1080 and 1010 Ma

Manoel S. D'Agrella-Filho [a,*], Igor I.G. Pacca [a], Ricardo I.F. Trindade [a], Wilson Teixeira [b], M. Irene B. Raposo [b], Tullis C. Onstott [c]

[a] *Instituto de Astronomia, Geofísica e Ciências Atmosféricas, Universidade de São Paulo, Rua do Matão 1226, São Paulo 05508-090, SP, Brazil*
[b] *Instituto de Geociências, Universidade de São Paulo, Rua do Lago 562, São Paulo 05508-080, SP, Brazil*
[c] *Department of Geological and Geophysical Sciences, Princeton University, Princeton, NJ 08544, USA*

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Abstract

Paleomagnetic results from unmetamorphosed Mesoproterozoic mafic dikes of Salvador, northern São Francisco craton, yield either an easterly direction with high upward inclination (Dm = 110.8°, Im = −79.3°, α_{95} = 8.4°, N = 8, paleopole at 122.7°E, 6.4°N), or a westerly direction with high downward inclination (Dm = 290.7°, Im = 64.7°, α_{95} = 7.5°, N = 7, paleopole at 282.6°E, 5.6°N). Additional paleomagnetic analyses on country rocks (granulites and amphibolitic dikes) yield paleomagnetic poles similar to those obtained in other metamorphic rocks from the São Francisco craton affected by the Transamazonian orogeny (2200–1900 Ma). Positive baked contact tests in two dikes, together with rock magnetic studies, supported by petrographic and radiometric analyses, indicate that the remanent directions isolated for the unmetamorphosed dikes are primary TRM's. $^{40}Ar/^{39}Ar$ step-heating analyses on biotite from a granulite sampled ∼90 cm away from the dike's contact yield an integrated age of 1021 ± 8 Ma, whereas the plagioclase from the dike itself yields an integrated age of 1003 ± 33 Ma, interpreted to be the time of the emplacement. Our data in conjunction with the available late Mesoproterozoic poles from the São Francisco craton allow to best constrain its APW path between 1080 and 1010 Ma.
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Keywords: Mafic dikes; Mesoproterozoic; São Francisco craton; Rodinia; Paleomagnetism; $^{40}Ar/^{39}Ar$ geochronology

1. Introduction

Since the seminal works of Dalziel (1991), Hoffman (1991) and Moores (1991) several contributions have attempted to decipher the Meso- to Neoproterozoic dynamics that gave birth to the Rodinia Supercontinent (e.g., see Meert and Powell, 2001, for a review). One of the key pieces in reconstructing Rodinia is the Congo craton (Africa) and its Brazilian counterpart, the São Francisco craton. Paleomagnetic-based reconstructions of Rodinia proposed by Weil et al. (1998), D'Agrella-Filho et al. (1998) and Dalziel et al. (2000) place the Congo/São Francisco craton south to southeastward from the (present-day) margin of Laurentia. In two of these reconstructions the Kalahari craton is placed between Laurentia and the Congo/São Francisco blocks (Fig. 1). However, some recent works en-

* Corresponding author. Fax: +55-11-3091-5034.
E-mail address: dagrella@iag.usp.br (M.S. D'Agrella-Filho).

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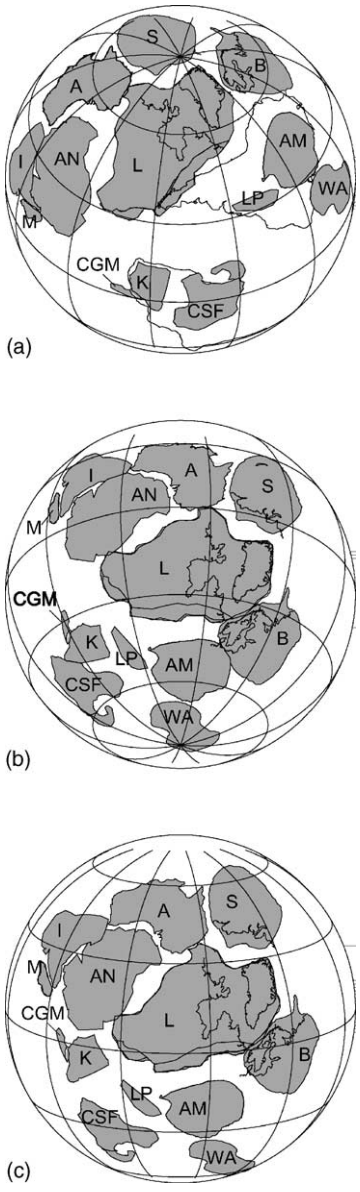


Fig. 1. Rodinia reconstructions (after Cordani et al., 2003). (a) Dalziel et al. (2000), (b) Weil et al. (1998) and (c) D'Agrella-Filho et al. (1998). Abbreviations: L, Laurentia; M, Madagascar; I, India; AN, East Antarctica; A, Australia; S, Siberia; CMG, Coats Land-Maudheim-Grunehogna; K, Kalahari; CSF, Congo/São Francisco; LP, Rio de la Plata; AM, Amazonia; B, Baltica; WA, West Africa. Reconstructions done with the GMAP package of T.H. Torsvik and M. Smethurst.

visage a large Neoproterozoic ocean (the Brasiliano Ocean) separating most of the South American and African units from the remainder fragments of Rodinia (e.g., Pimentel et al., 1999; Kröner and Cordani, 2003; Cordani et al., 2003; Pisarevsky et al., 2003).

The position of the Congo/São Francisco craton at late Mesoproterozoic times has been constrained mainly by paleomagnetic and geochronological data from the Ilhéus, Olivença and Itaju do Colônia tholeiitic dike swarms. This data set defines an apparent polar wander (APW) path for the São Francisco craton between 1080 and 1010 Ma (D'Agrella-Filho et al., 1990; Renne et al., 1990). Here, we present new paleomagnetic data obtained for geochemically similar dikes that crop out 400 km to the north of Ilhéus and Olivença, around the city of Salvador (Bellieni et al., 1998) (Fig. 2). Also shown are paleomagnetic analyses on amphibolitic dikes and granulites (country rocks), and a baked contact test. Additional $^{40}Ar/^{39}Ar$ ages (dike and country rocks) are also presented. The new APW path for the São Francisco craton that results from the inclusion of the high-quality Salvador paleopoles and the implications for the Rodinia configuration are discussed.

2. Geological setting and sampling

The study area lies in the north-eastern part of the São Francisco craton (Fig. 2a) (e.g., de Brito Neves and Cordani, 1991), which was dominantly formed/reworked during the Transamazonian orogeny (2200–1900 Ma), mirrored by the Salvador-Curaçá and Itabuna-Atlantic Coast belts (e.g., Teixeira and Figueiredo, 1991; Ledru et al., 1994; Teixeira et al., 2000). Such a tectonic framework results from continental shortening and collision of several Archean blocks (e.g., Gavião, Jequié, Serrinha, North Gabon, Chaillu) (e.g., Teixeira et al., 2000). The Contendas-Jacobina lineament (CJL in Fig. 2a) represents the front of the Transamazonian collision which led to the emplacement of a string of granitoid and sienitic plutons (not shown in Fig. 2a).

The studied rocks comprise amphibolites and unmetamorphosed mafic dikes that crop out along the coast of Salvador (Fig. 2b). The amphibolites crop out as boudinated dikes intrusive in the medium- to high-grade rocks of the Paleoproterozoic Salvador-Curaçá

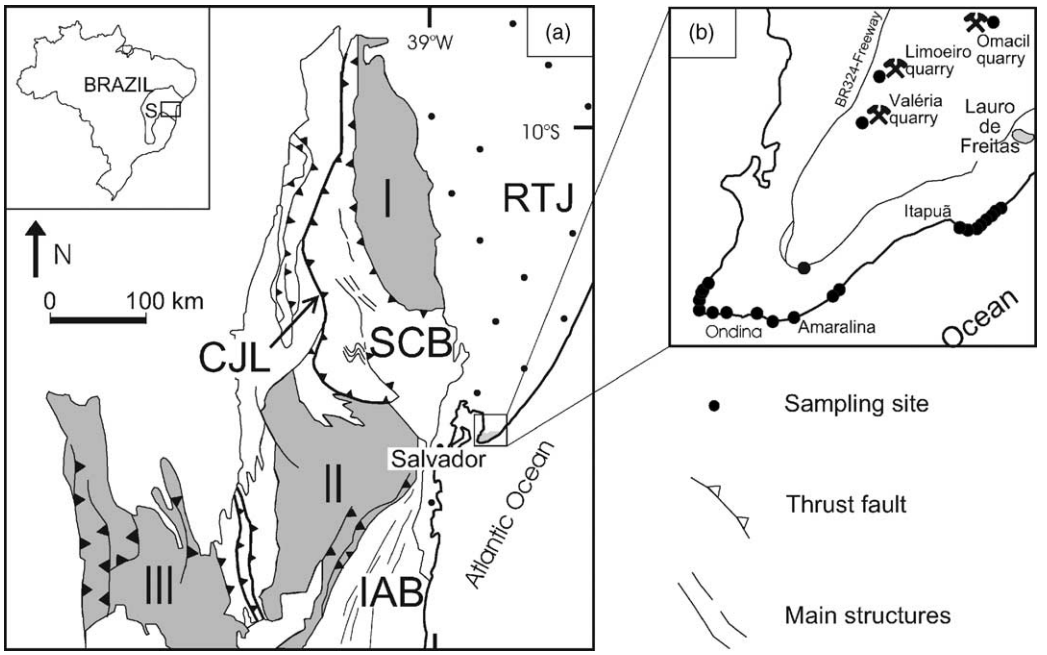


Fig. 2. (a) Tectonic framework of northern São Francisco craton (after Teixeira et al., 2000). Archean blocks: I, Serrinha Block; II, Jequié Block; III, Gavião Block. Paleoproterozoic mobile belts: SCB, Salvador-Curaçá Belt; IAB, Itabuna-Atlantic Coast Belt; CJL, Contendas-Jacobina lineament; RTJ, Cretaceous Recôncavo-Tucano-Jatobá Basin; S, São Francisco craton (inset). (b) Location of sampling sites (●) in the Salvador region.

belt (e.g., Teixeira et al., 2000). Chemically, they are latites to lati-andesites, of calc-alkaline affinity, composed of plagioclase (andesine), hornblende, biotite, opaque minerals and rare clynopyroxene. Zircon and apatite are the main accessory minerals (Moraes-Brito et al., 1989). Radiometric ages (mostly U–Pb in zircons) between 2200 and 2000 Ma for the syntectonic granitic plutons and host granulite rocks indicate the time of reworking and regional metamorphism, and give also a time-constraint for the amphibolitic dikes (e.g., Mascarenhas and Garcia, 1989; Sabaté et al., 1994; Rosa et al., 1999; Ledru et al., 1994).

The unmetamorphosed dikes (tholeiites) cut across both the amphibolitic dikes and granulites. They range from a few centimeters up to 50 m wide, and are composed of fresh plagioclase, augite and opaque minerals, but hypersthene and pigeonite may be also present (Moraes-Brito et al., 1989; Bellieni et al., 1998). The K–Ar ages for these dikes ranging from 1300 to 1730 Ma, are interpreted to be due to excess argon (e.g., Parenti Couto et al., 1983). A younger U–Pb baddeleyite age of 924 ± 24 Ma (Heaman, 1991) which defines the emplacement age of a dike in Ondina (site SA22, Table 1) supports this interpretation.

Paleomagnetic analyses were performed in 72 oriented hand samples and 7 cylindrical cores from 23 unmetamorphosed mafic dikes located along the coast of Salvador (16 dikes), and at the Omacil (1 dike) and Valéria (6 dikes) quarries (Fig. 2b). In the same area, 111 oriented hand samples were collected from 14 amphibolitic dikes (43 samples) and from granulitic rocks (68 samples) at 7 sites for baked contact test. Three to four block samples spaced at least one half meter apart were collected from each dike (considered as a sampling site). Whenever possible, both solar and magnetic compasses were used for orienting samples. For the majority of the samples the agreement of these measurements was good, the difference between them being generally less than 6°.

A sample (JA51) from a 30 m thick unmetamorphosed dike (site SA17, Table 1) and three samples (granulites) from the country rock were collected at different distances from this dike (JA57 at 0.90 m,

Table 1
Paleomagnetic results from the Salvador area

Site	Samples	*N*	Site mean directions				VGP		Rotated VGP pole	
			Dec (°)	Inc (°)	α_{95} (°)	*K*	P_{long} (°E)	P_{lat} (°N)	P_{long} (°E)	P_{lat} (°N)
Unmetamorphosed dikes										
SA1[a] C	BH756–758	10	178.8	59.6	12.4[b]		323.6	−62.2	320.1	−62.1
SA2[a] B	BH721–723; BH744–746; BH749–750	6	91.1	−49.9	5.8	135	81.9	5.6	82.1	3.9
SA3[a] B	BH724	1	112.3	−21.9			68.4	−18.6	68.9	−20.5
SA4[a] B	BH725–726	2	118.5	−32.7	16.4	234	77.1	−22.1	77.7	−23.9
SA5[a] B	BH738; BH741–743	3	95.1	−36.2	22.2	32	72.3	−0.3	72.6	−2.2
SA7[a] B	BH751–752	2	101.7	−37.4	20.8	146	74.7	−6.1	75.1	−8.0
SA8 A	BH132–135; BH142	5	295.9	57.1	5.5	195	275.3	11.4	275.9	12.9
SA9 A	BH143–146	4	286.9	59.6	4.9	340	274.8	3.9	275.2	5.4
SA10 C	BH700–703	2	151.0	74.9	29.4	74	338.3	−37.1	337.1	−37.6
SA12 A	BH153–155	1	74.2	−76.2			114.9	18.5	114.6	17.5
SA14 A	BH216–219	2	305.2	−86.8	20.3	153	146.8	16.5	146.5	16.6
SA15 A	BH221–222	2	98.3	−72.9	16.4	233	110.1	6.7	110.2	5.6
SA16 A	BH194–196	3	293.2	71.6	15.0	68	290.9	1.4	291.2	2.5
SA17[a] A	BH201–204; JA36–42	6	86.8	−80.5	4.7	206	122.5	13.3	122.4	12.6
SA18[a] A	BH766–768	3	136.8	−64.6	3.6	1149	111.6	−19.0	112.5	−20.1
SA19[a] A	BH769–771	3	149.7	−65.0	14.5	73	119.3	−24.2	120.4	−25.0
SA20[a] A	BH772–773	2	303.9	72.3	14.3	304	293.5	6.8	294.0	7.8
SA21[a] A	BH774–776	2	302.7	75.1	8.8	802	298.2	2.7	298.6	3.6
SA22 C	BH172–175	4	167.2	61.4	13.2	49	339.7	−58.4	336.8	−58.9
SA23 A	BH162–164	3	82.5	−83.5	8.2	224	128.3	14.3	128.1	13.8
Granulites										
SA8b A	BH136–141 (138)	1	287.8	50.2			266.1	8.0	266.6	9.7
	BH136–141 (141)	1	293.2	−38.2			213.8	26.1	213.7	28.0
SA11b	BH705–708	2	68.4	57.8	35.6	52	9.0	8.2	9.5	6.8
SA16b A	BH197–200 (197, 198, 199)	3	274.5	65.1	11.2	122	278.9	4.5	279.3	5.9
	BH197–200 (200)	1	28.8	57.1			348.4	32.5	349.8	31.7
SA17b A	BH205–215; BH620–634; BH1E–10E	15	50.8	−82.5	5.1	58	129.2	22.0	128.7	21.5
Amphibolite dikes										
SA24	BH753–755	2	275.9	−55.9	34.1	34	196.5	12.2	196.5	13.8
SA25	BH760–762	3	283.3	−37.1	10.2	145	213.5	16.7	213.6	18.6
SA26	BH747–748	1	313.0	−35.3			216.7	44.5	216.4	46.4
SA28	BH715–717	1	8.3	54.6			330.7	41.3	332.7	41.0
SA31	BH734–736	2	285.0	−22.3	31.6	65	223.2	16.9	223.4	18.9
SA32	BH737	1	324.1	−56.3			188.3	49.9	186.8	51.3
SA33	BH150–152	2	278.0	−32.4	32.2	62	215.9	11.4	216.1	13.3
SA34	BH156–158; BH161	1	302.1	−59.6			191.6	32.7	191.0	34.2
SA35	BH223–226	3	3.9	61.2	23.2	29	325.0	34.5	326.6	34.4
SA37	BH763–765	3	307.7	−36.0	20.3	38	216.2	39.6	216.0	41.5

N, number of samples used in site mean directions; Dec, declination; Inc, inclination; α_{95}, Fisher's semi-angle confidence cone; K, Fisher's precision parameter; VGP, virtual geomagnetic pole; P_{long}, pole longitude; P_{lat}, pole latitude; A, component A; B, component B; C, component C (see text). Sites SA2–SA7 were collected in the Valéria quarry. Rotated VGP: virtual geomagnetic pole position after 2° clockwise rotation about the rotation pole at 36°04′W; 8°11′S (Milani and Davison, 1988).

[a] Sites analyzed in Princeton University.

[b] MAD (maximum angular deviation; Kirschvink, 1980).

JA58 at 7.6 m and JA60 at 24 m) for $^{40}Ar/^{39}Ar$ determinations.

3. Magnetic mineralogy

The magnetic mineralogy of the unmetamorphosed dikes and country rocks (granulites and amphibolitic dikes) was extensively studied. Thermomagnetic curves in air were obtained for at least one sample per site, using a horizontal thermomagnetic balance (Fig. 3). These results are compared to typical thermal demagnetization curves (Fig. 4). Magnetic fields up to 1.0 T were applied to the samples to produce isothermal remanent magnetization (IRM) acquisition curves (Fig. 5a). Hysteresis loops for at least two samples per site were obtained in a Molspin vibrating sample magnetometer (Fig. 5b–f). They give bulk coercive force, Hc, saturation magnetization, Js, and saturation isothermal remanence, Jrs, after subtraction of the paramagnetic contribution from the high field portion of the curve. Coercivity of remanence (Hcr) was also determined using the same instrument. Then, Jrs/Js and Hcr/Hc ratios were calculated. Additionally, optical and scanning electron microscopy (SEM) were used to characterize the magnetic carriers of the unmetamorphosed dikes.

3.1. Unmetamorphosed dikes (UD)

Most UD are characterized by high magnetic susceptibilities (0.001–0.1, SI units) and NRM intensities (0.023–18.1 A/m). Thermomagnetic curves are typical of Ti-poor titanomagnetite (Fig. 3a), with Curie temperatures between 570 and 587 °C. Correspondingly, laboratory unblocking temperatures (UT) for most of the samples are bracketed within 500 and 600 °C, even though some samples comprise grains with UT lower than 400 °C (Fig. 4). Almost all samples show a "hard-shoulder" demagnetization path suggesting that remanence resides in hard coercivity fractions (Dunlop and Özdemir, 2000). The samples with lower UT present a small decrease in saturation magnetization in the thermomagnetic curves (Fig. 3b). This behavior suggests the presence of maghemite, which changes to hematite during heating at temperatures between 330 and 365 °C, giving smaller magnetization in the cooling curve.

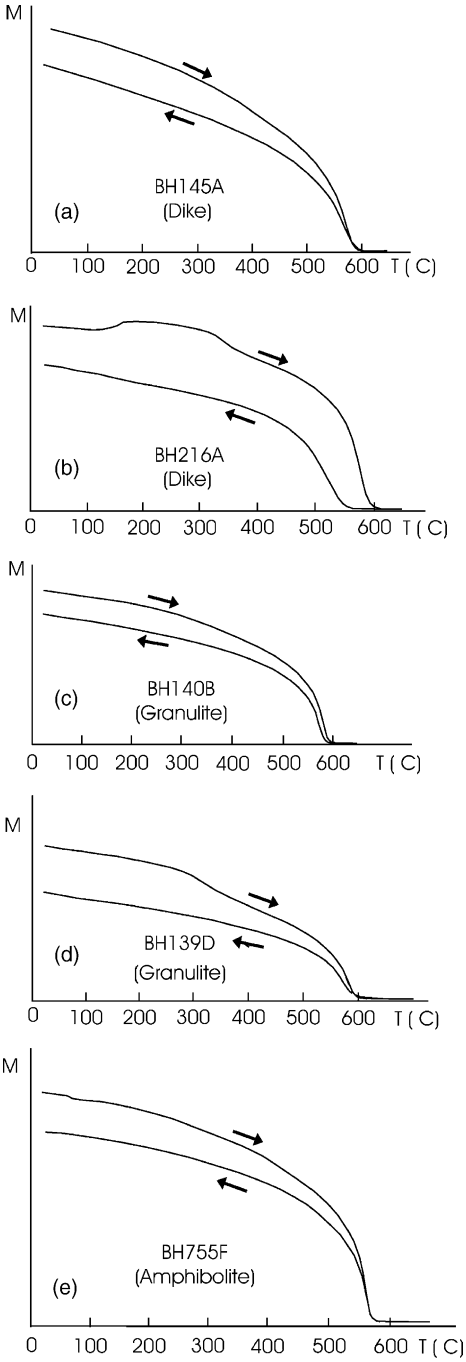


Fig. 3. Saturation magnetization versus temperature curves obtained for unmetamorphosed dikes (a and b), granulites (c and d), and amphibolites (e).

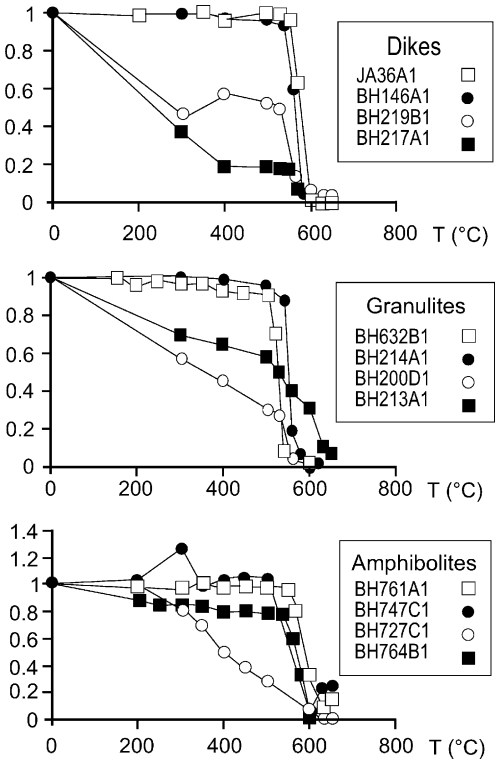


Fig. 4. Normalized intensities versus temperature for representative specimens from the unmetamorphosed dikes, granulites and amphibolites, obtained during thermal demagnetization.

Saturation of remanence (Jrs) at fields between 135 and 300 mT (Fig. 5a) and coercivities (Hc) between 9.5 and 76 mT are also typical of titanomagnetites (Day et al., 1977). A typical example of hysteresis loop for these rocks is shown in Fig. 5b, before subtraction of the paramagnetic contribution. Hysteresis parameters for most UD fall into the pseudo-single domain field of the Day diagram (Fig. 6).

In the polished thin sections magnetite occurs as coarse grains intergrown with ilmenite, the later commonly occurring as either coarse patches or relatively thin lamellae. These features are inferred to result from high-temperature oxidation/exsolution upon initial cooling of the dikes (Haggerty, 1976).

3.2. Granulites and amphibolitic dikes

Granulitic rocks show NRM intensities between 4×10^{-4} and 7.7 A/m, and magnetic susceptibilities varying from 80×10^{-6} up to 13×10^{-2} SI. The amphibolitic dikes display NRM intensities ranging from 6×10^{-4} up to 6 A/m, whereas magnetic susceptibilities are between 352×10^{-6} and 19×10^{-2} SI. Thermomagnetic curves for granulite samples present Curie temperatures between 565 and 587 °C, typical of Ti-poor titanomagnetites (Fig. 3c). Some samples also show a small inflection around 350 °C that can be ascribed to maghemite (Fig. 3d). Laboratory UT are between 500 and 600 °C (Fig. 4b). Note that some samples have shown long demagnetization tails during thermal treatment suggesting soft magnetite grains as the main carriers of remanence (Dunlop and Özdemir, 2000). Samples collected far from unmetamorphosed dikes typically have narrow-waisted histeresis loops (Fig. 5c), with coercivities ranging between 1.5 and 5.5 mT, occupying the MD field in the Day plot (Fig. 6). By contrast, samples collected at less than 2 m from the dike SA17 have coercivities usually higher than 6.0 mT (Fig. 5d) and fall into the pseudo-single domain (PSD) field (Fig. 6).

The amphibolitic dikes yield Curie temperatures between 555 and 587 °C (Fig. 3e), UT between 500 and 600 °C (Fig. 4c), fast saturation of IRM between 120 and 250 mT (Fig. 5a), and coercivities ranging from 1 to 4.5 mT (Fig. 5e), which are typical of Ti-poor titanomagnetites. Some samples, however, are weakly magnetic and show saturation magnetizations below the sensitivity of our magnetic balance. Also, their hysteresis loops are strongly linear pointing to the predominance of paramagnetic minerals (Fig. 5f). All samples fall into the multidomain (MD) field in the Day plot (Fig. 6).

4. Paleomagnetism

Three to four specimens 2.5 cm in diameter and height were cut from cores extracted from each hand sample. These samples were then submitted to both alternating field (AF) and thermal demagnetization to isolate magnetic components and test its stability. Spinner magnetometers (Molspin and Schonstedt SM-1) were used for remanent magnetization measurements.

A group of unmetamorphosed dikes (Table 1) has been analyzed at Princeton University (USA) in 1989. The other samples were analyzed at the paleomag-

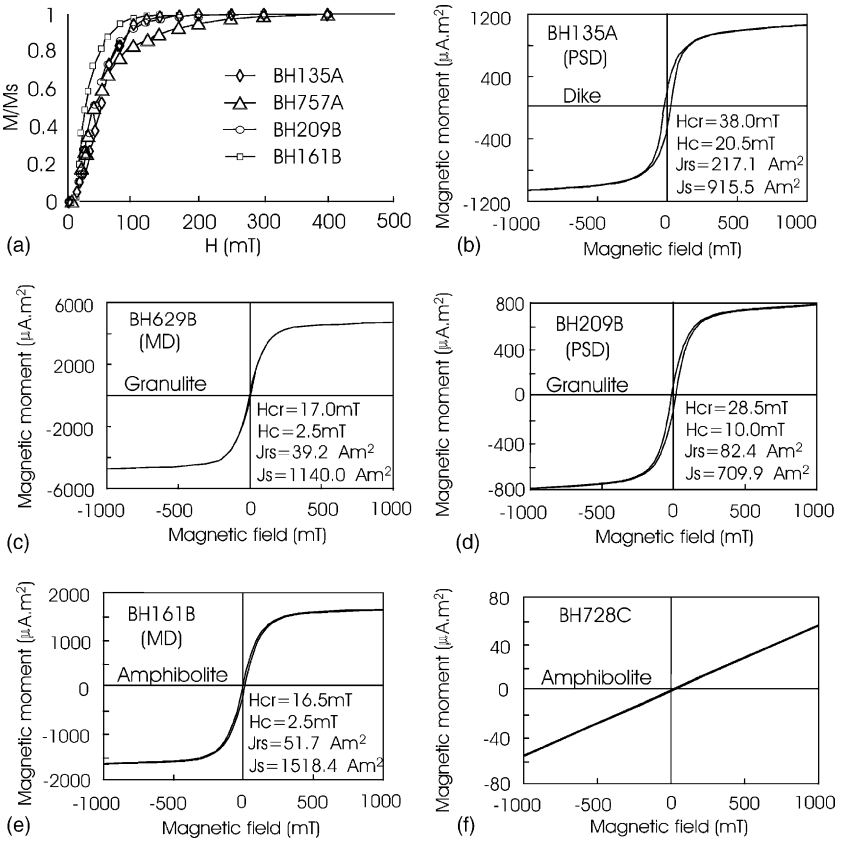


Fig. 5. (a) Normalized IRM acquisition curves obtained for the unmetamorphosed dikes (samples BH135A and BH757A), granulites (samples BH209B) and amphibolites (BH161B). Hysteresis loops obtained for unmetamorphosed dikes (b) granulites (c and d) and amphibolite dikes (e and f).

netic lab of the University of São Paulo. Principal component analysis (Kirschvink, 1980) was applied to determine the characteristic magnetization direction for most sites, but for site SA1 the characteristic magnetization was identified by the converging remagnetization circles method (Halls, 1976, 1978). The directions reported in Table 1 represent the average of sample directions yielding the best cluster after combined AF and thermal demagnetization. Site-mean directions and paleomagnetic poles were determined using Fisher (1953) statistics.

4.1. Unmetamorphosed dikes

Three groups (A, B and C) of characteristic magnetization (ChRM) directions (Fig. 7a) could be isolated after elimination of secondary components at fields up to 10–20 mT. Component A was obtained from most of the analyzed dikes (Table 1). It corresponds to a western declination, steep downward (A_R component) inclination, magnetization or to an eastern declination, steep upward (A_N component) inclination, magnetization (Fig. 7a). The AF demagnetization for specimen BH772A1 (site SA20) isolating component A_R is shown in Fig. 8.

Component B was isolated after AF demagnetization of samples from five dikes sampled at Valéria quarry (Fig. 7a). It corresponds to an east to southeast declination, moderate upward inclination magnetization. An example of AF demagnetization isolating this component is shown for the specimen BH724A1 in Fig. 8 (site SA3). Component C was obtained after AF and thermal demagnetization of samples from dikes SA1, SA10 and SA22 (Table 1). These samples present

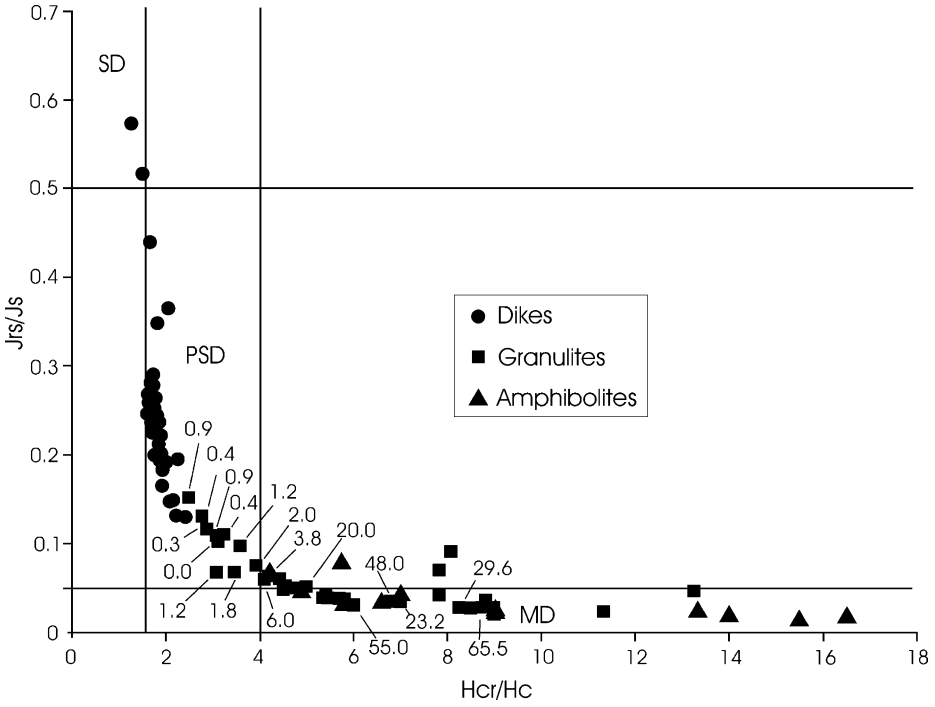


Fig. 6. Day plot. Numbers in granulite samples indicate the distance from dike's contact in meters. Abbreviations: SD, single-domain, PSD, pseudo-single domain; MD, multidomain.

a south to southeast declination, steep downward inclination, magnetization (Fig. 7a). Fig. 8 shows the AF demagnetization for specimen BH700A1 (site SA10), where this component was isolated. A stable magnetization could not be defined for samples from site SA1. However, different remagnetization circles were defined by AF demagnetization in these rocks (Fig. 9) so that the converging remagnetization circle's method of

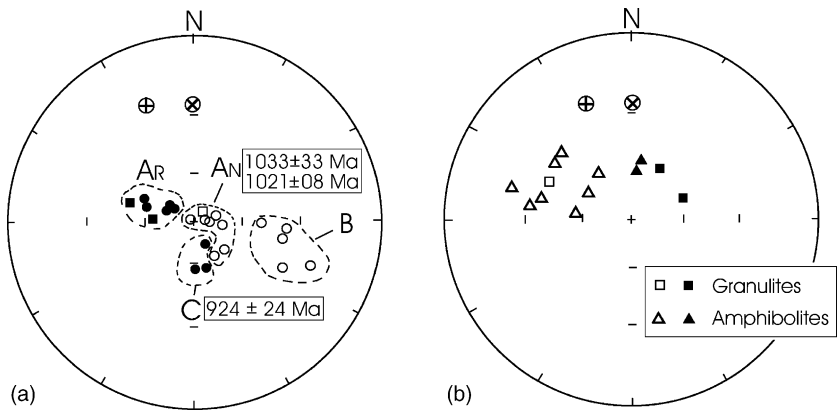


Fig. 7. Site mean directions. (a) Unmetamorphosed dikes (circles) and granulites (squares) close to the contact, (b) granulites (squares) and amphibolites (triangles) far from unmetmorphosed dikes. Closed dashed lines indicate each group of directions found in the Salvador area. $^{40}Ar/^{39}Ar$ and U–Pb datings obtained for two dikes are also indicated for components A_N and C. Open (full) symbols represent upward (downward) inclinations. The symbols (⊕) and (⊗) are the present geomagnetic and dipolar fields, respectively. Equal-area stereographic projection.

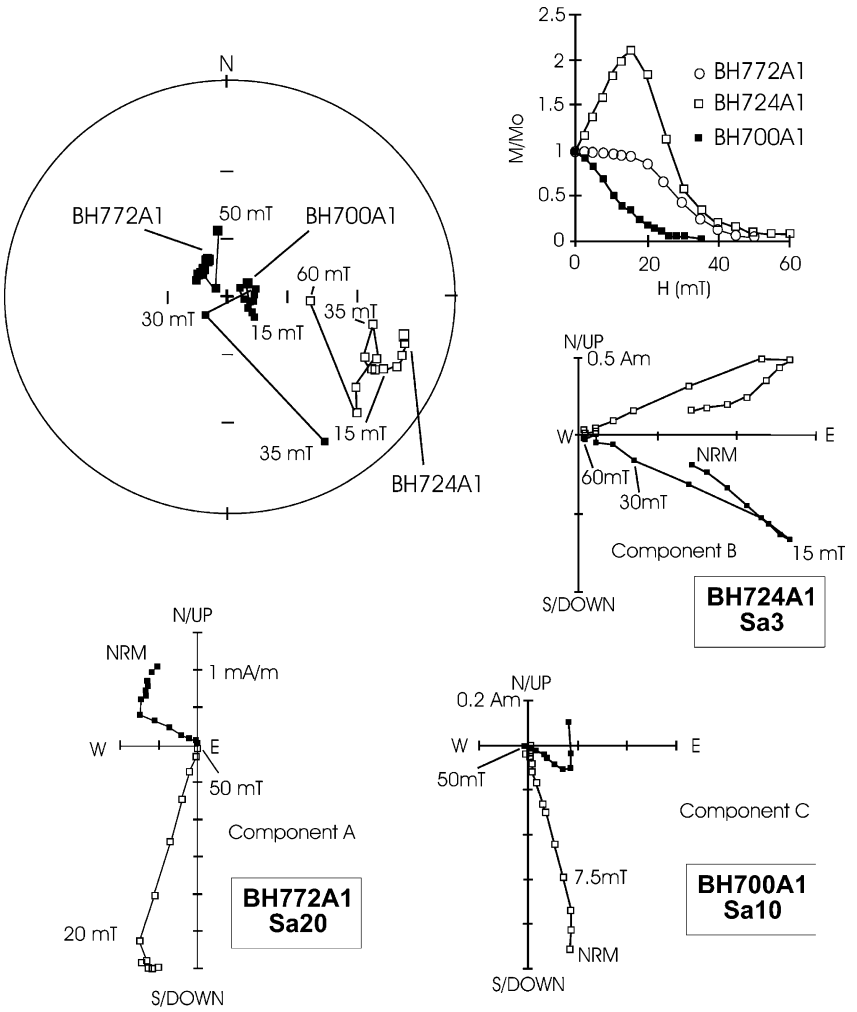


Fig. 8. Magnetic behavior of unmetamorphosed dikes: specimens BH772A1 (component A), BH724A1 (component B) and BH700A1 (component C). In the stereographic projections open/full symbols represent upward/downward inclinations. Corresponding normalized intensity curves and orthogonal diagrams (full/open symbols are horizontal/vertical plane projections) are also shown.

Halls (1976, 1978) could be used to isolate component C. Principal component analysis was applied to 10 demagnetization paths defining great circles (e.g., specimens BH756A1, BH756C1 and BH757C1 in Fig. 9a). The pole of the best plane defined by the normal to these great circles (Fig. 9b) gives the characteristic direction for site SA1.

4.2. Granulites and amphibolitic dikes

Although several samples from the country rocks have presented either erratic demagnetization patterns or very low magnetization intensities (below the noise level of the spinner magnetometer), two groups of characteristic magnetization directions were disclosed (Fig. 7b). AF demagnetization was efficient to isolate a characteristic northwest directed, upward inclination, magnetization in 8 out of 14 amphibolite dikes (the AF demagnetization pattern for specimen BH747C1 (site SA26) is given as an example in Fig. 10). Samples from two other amphibolitic dikes yield a north to northeast declination, steep positive inclination, magnetization (Fig. 7b) after combined AF and thermal demagnetization. Demagnetization applied to speci-

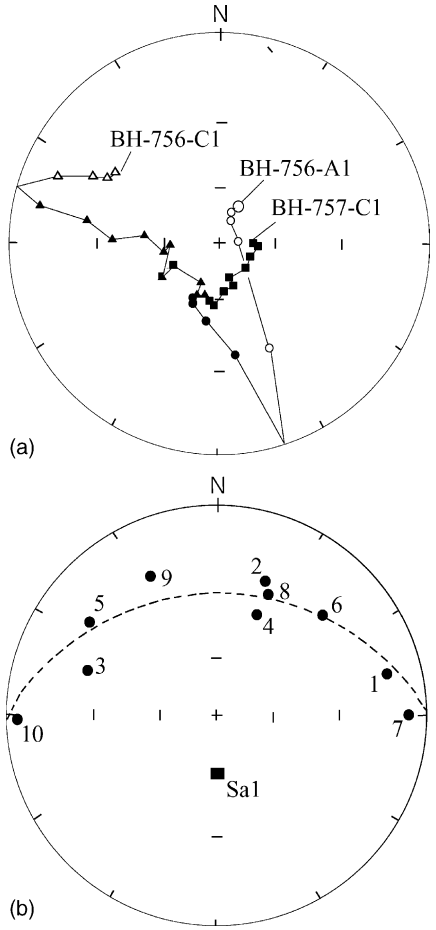


Fig. 9. Magnetic behavior of samples from site SA1. (a) Specimens BH756A1 (circles), BH756C1 (triangles) and BH757C1 (squares). (b) Remagnetization circle obtained for the 10 specimens from site SA1. Point SA1 represents the characteristic direction for site SA1. Symbols are the same as in Fig. 8.

men BH225C1 from one of these dikes (site SA35) is shown in Fig. 10.

Granulites collected far from the unmetamorphosed dikes have shown magnetic directions similar to those disclosed for the amphibolitic dikes (see Table 1). This was the case for one sample from site SA8b, located at 5 m from the contact of a ~3 m thick dike and for a sample from site SA16b, collected at 1.7 m from the contact of a ~1.5 m thick dike. Demagnetization performed on specimens BH141A1 (site SA8b) and BH200B1 (site SA16b) is shown in Figs. 10 and 11, respectively. After thermal demagnetization a similar direction was also disclosed for samples collected far from any dike (site SA11b), to establish the characteristic direction of the country rocks. The demagnetization behavior of specimen BH706B1 is given as an example in Fig. 11.

5. Baked contact test

Unmetamorphosed dikes and country rocks (granulites and amphibolitic dikes) far from them show different characteristic magnetizations (Fig. 7). However, a parallelism of directions between the dike and the host rock is expected close to the contact if dike's magnetization is thermoremanent (TRM) in origin. In this way, 68 samples from granulitic rocks were collected close to six unmetamorphosed dikes (sites SA8b, SA9b, SA16b, SA17b, SA22b and SA23b) to investigate the primary character of their characteristic magnetization. Most of these samples (35) belong to site SA17b (Table 1) and were collected near to a 30 m thick dike.

Samples from sites SA9b, SA22b and SA23b showed unstable demagnetization behavior, and no coherent magnetization direction could be isolated for these sites. The magnetic behavior of the country rocks from the remainder three sites provides a positive baked contact test as shown by the examples of Fig. 11. The paleomagnetic results obtained for these three sites are summarized bellow.

5.1. Site SA16b

After eliminating viscous components at 10–15 mT, the thermal demagnetization of samples from the dikes's contact (BH197, 198, 199) yields a west to northwest declination, positive inclination magnetization (e.g., specimen BH197A1 in Fig. 11). This component is similar to the magnetization found for the dike itself (specimen BH196C1 in Fig. 11). A direction characteristic of the country rocks was disclosed after thermal treatment on two specimens of sample BH200, collected 1.7 m far from the dike (specimen BH200B1 in Fig. 11).

5.2. Site SA8b

Four samples (BH136, BH137, BH139 and BH140) from this site yield unstable demagnetization behavior.

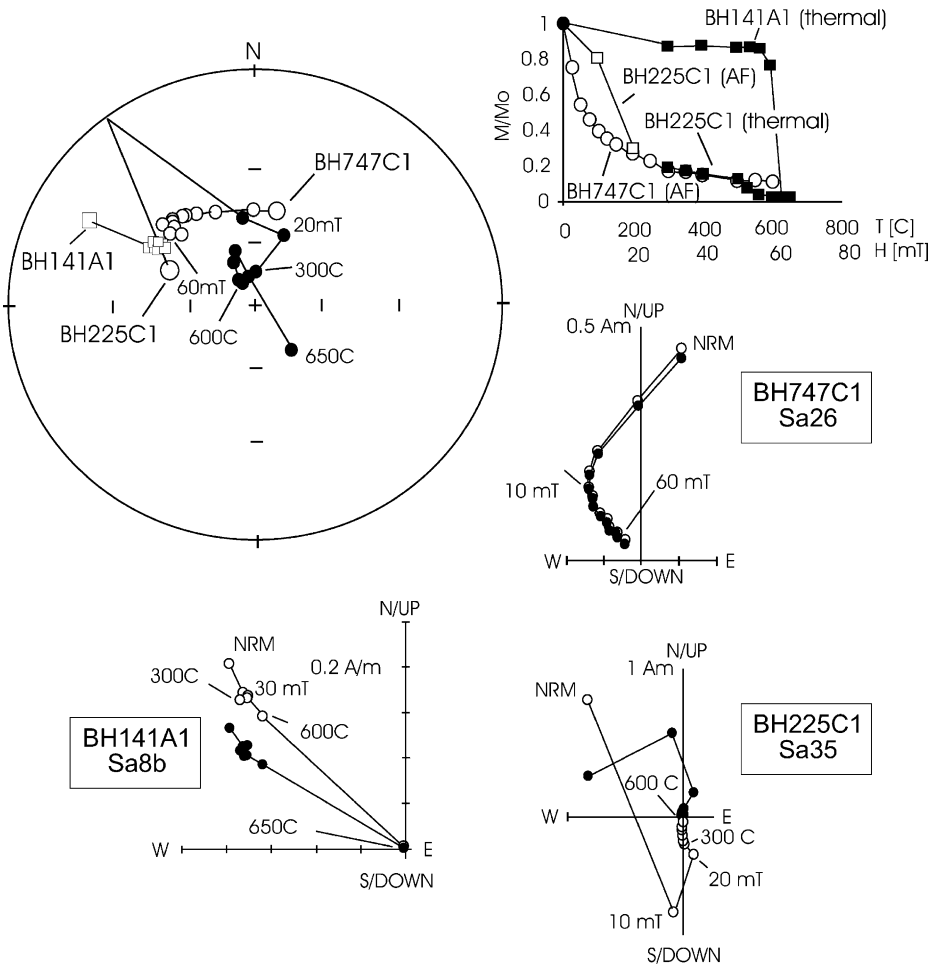


Fig. 10. Magnetic behavior of amphibolite dikes BH747C1 (upward inclination) and BH225C1 (downward inclination) and a granulite (BH141A1) located at 5 m from the contact of a ~3 m thick dike. Symbols are the same as in Fig. 8.

AF and thermal demagnetization of sample BH138, from the dike's contact, yielded a west declination, steep positive inclination. This magnetization is similar to that found for the dike itself (site SA8 in Table 1). Sample BH141, sampled ~5 m away from the dike, yields a northwest declination, moderate negative inclination, magnetization after AF and thermal demagnetization, which is characteristic of the country rocks (see specimen BH141A1 in Fig. 10).

5.3. Site SA17b

The closest 16 samples to the dike's contact but sample BH10 yield a north to northeast declination, steep negative inclination, magnetization after AF and thermal demagnetization (see specimen BH632D1 in Fig. 11). This component is similar to that found for the dike (~30 m thick) itself (specimen JA37 in Fig. 11). However, samples collected more than 20 m far from dike's contact yield an erratic demagnetization pattern.

The characteristic magnetization of the country rocks was disclosed for granulites sampled far from the unmetamorphosed dikes (see above). Their corresponding virtual geomagnetic poles, together with those obtained from the amphibolite dikes, are comparable to the results obtained for other basement rocks from Congo/São Francisco in a pre-

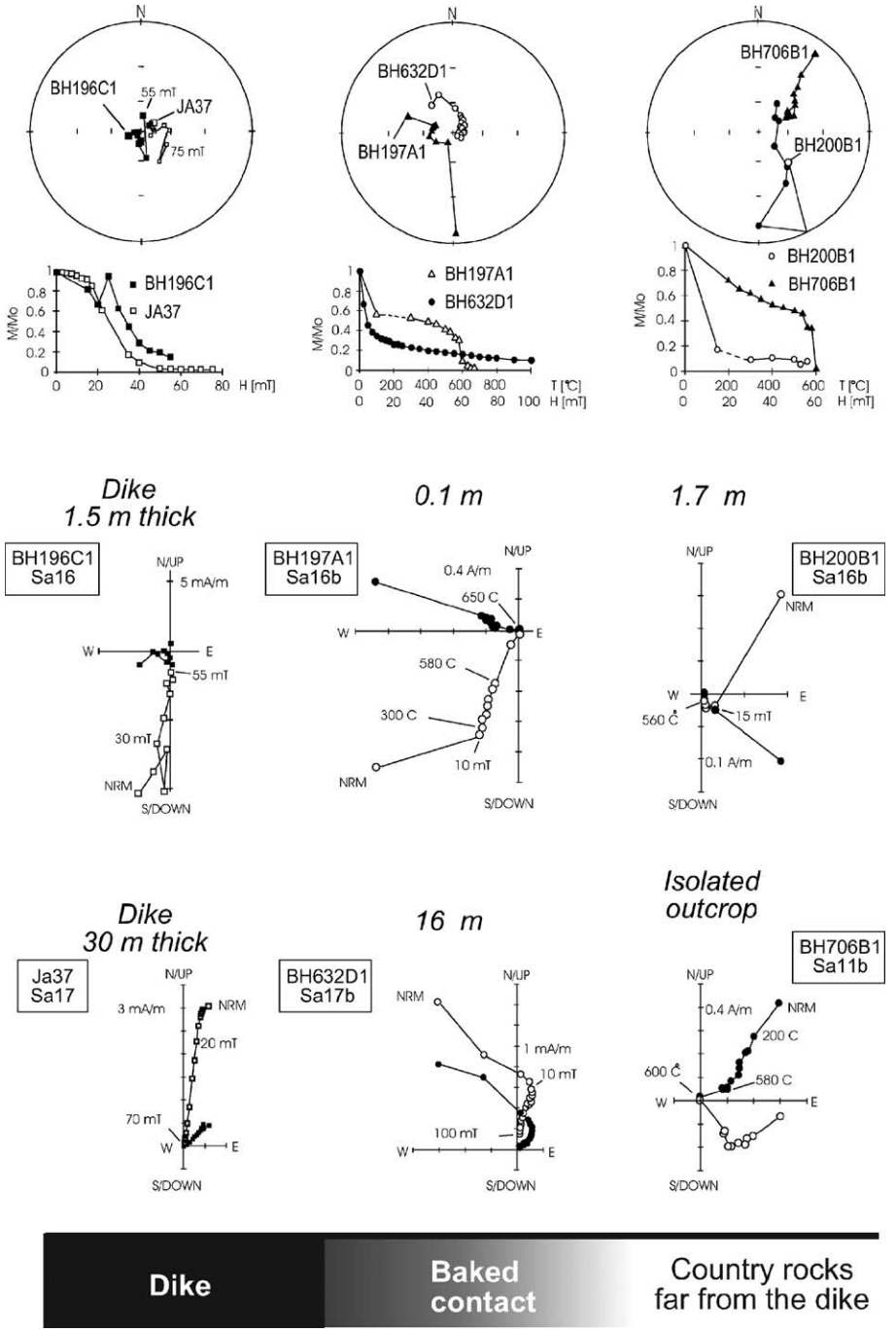


Fig. 11. Baked contact test (sketch not in scale): magnetic behavior of unmetamorphosed dikes (specimens BH196C1 and JA37) yielding component A, of granulites at the contact with the dike (specimens BH197A1 and BH632D1) yielding component A, and of granulites far from the dike (specimens BH220B1 and BH706B1) yielding the characteristic magnetization of the country rocks. Distances from the contact are indicated in meters. Symbols are the same as in Fig. 8.

Table 2
$^{40}Ar/^{39}Ar$ analytical data

Temperature (°C)	^{36}Ar	^{37}Ar	^{38}Ar	^{39}Ar	^{40}Ar	Age (Ma)	1 S.D.
Sample JA51							
100	9.72E−01	6.51E−00	5.53E−00	2.24E+01	6.40E+02	936.57	7.39
200	1.55E−01	3.91E−00	2.40E−00	7.85E−00	4.49E+02	1487.08	12.94
300	3.76E−01	2.16E+01	9.84E−00	1.90E+01	1.84E+03	2131.97	10.26
400	7.75E−01	5.51E+01	3.69E+01	7.00E+01	4.66E+03	1708.99	3.89
500	4.34E−00	1.79E+02	1.24E+02	3.20E+02	1.21E+04	1099.97	2.25
600	6.09E−01	9.53E+01	7.92E+01	2.17E+02	7.41E+03	1092.74	1.53
700	3.55E−01	4.44E+01	3.57E+01	1.06E+02	3.51E+03	1059.51	1.77
800	3.67E−01	5.77E+01	7.52E+01	2.04E+02	6.36E+03	1022.13	1.07
900	2.14E−01	1.95E+01	1.80E+01	5.48E+01	1.78E+03	1041.04	3.59
1000	1.51E−01	2.95E+01	3.21E+01	1.05E+02	3.28E+03	1030.27	2.03
1100	6.65E−02	2.57E+01	2.59E+01	8.38E+01	2.63E+03	1035.71	2.72
1400	1.86E−01	4.40E+01	2.77E+01	1.05E+02	3.37E+03	1047.62	2.07
Sample JA57							
500	6.19E+01	2.61E+02	3.07E+02	7.91E+02	3.92E+04	865.83	49.90
598	1.84E+01	2.94E+02	2.23E+02	1.60E+03	7.43E+04	1249.84	20.79
644	3.29E+01	2.19E+02	7.47E+02	5.91E+03	2.50E+05	1216.74	8.54
699	4.31E+01	3.03E+02	2.75E+03	1.93E+04	7.17E+05	1107.13	8.30
751	2.02E+01	3.79E+02	4.04E+03	2.89E+04	9.55E+05	1023.03	7.16
800	2.55E−01	3.46E+02	2.64E+03	2.00E+04	6.66E+05	1033.99	8.05
855	1.02E+01	2.74E+02	2.34E+03	1.70E+04	5.33E+05	984.6	4.76
895	2.05E−00	2.14E+02	2.27E+03	1.76E+04	5.76E+05	1019.36	6.65
946	1.96E+01	2.01E+02	2.01E+03	1.50E+04	4.91E+05	1008.96	18.06
996	2.51E+01	1.54E+02	1.39E+03	1.04E+04	3.46E+05	1015.95	23.67
1047	4.92E+01	1.73E+02	1.40E+03	1.06E+04	3.65E+05	1030.82	12.39
1097	7.20E+01	3.26E+02	2.20E+03	1.58E+04	5.79E+05	1079.74	15.41
1150	8.88E+01	2.35E+02	3.94E+03	3.00E+04	1.05E+06	1050.22	5.13
1195	1.23E+02	1.81E+02	5.05E+03	3.72E+04	1.29E+06	1042.36	5.12
Sample JA58							
500	1.29E+02	3.17E+07	1.34E+03	1.28E+03	7.59E+04	941.48	38.54
595	3.06E+01	1.72E+02	4.57E+01	2.37E+03	1.14E+05	1279.85	37.62
646	4.21E+01	1.24E+02	1.00E+03	9.05E+03	5.16E+05	1498.45	7.32
697	4.36E+01	1.93E+02	1.58E+03	1.00E+04	6.67E+05	1664.69	92.52
746	1.41E+02	1.21E+03	4.96E+03	4.45E+04	2.28E+06	1394.31	2.73
797	3.73E+01	4.25E+02	4.06E+03	3.34E+04	1.76E+06	1438.28	2.30
825	1.82E+01	2.26E+02	1.71E+03	1.49E+04	6.73E+05	1287.47	21.73
875	3.57E+01	4.71E+02	2.67E+03	2.27E+04	1.10E+06	1347.98	6.32
946	2.48E+01	3.83E+02	2.83E+03	2.20E+04	1.17E+06	1444.12	11.55
997	4.52E+01	1.02E+03	1.94E+03	1.55E+04	7.51E+05	1344.93	1.25
1046	5.75E+01	8.73E+02	1.50E+03	1.34E+04	6.39E+05	1319.91	2.90
1099	8.42E+01	3.17E+02	2.89E+03	2.40E+04	1.20E+06	1367.71	34.64
1125	9.13E+01	1.90E+02	2.78E+03	2.55E+04	1.29E+06	1379.02	25.19
1150	1.30E+02	1.60E+02	3.67E+03	3.33E+04	1.79E+06	1440.15	3.07
1175	1.34E+02	1.33E+02	2.89E+03	2.65E+04	1.48E+06	1472.59	6.74
1200	1.47E+02	6.75E+01	7.10E+02	6.26E+03	3.84E+05	1473.81	12.58
Sample JA60							
525	8.69E−00	2.84E+02	4.91E+02	1.62E+03	8.25E+04	1072.82	29.93
600	3.30E+01	1.67E+02	3.22E+02	2.36E+03	1.47E+05	1543.91	15.71
650	6.51E+01	1.45E+02	1.09E+03	8.27E+03	4.76E+05	1489.81	26.94
675	3.14E+01	1.33E+02	1.46E+03	1.08E+04	6.24E+05	1518.11	21.41
700	3.26E+01	1.42E+02	2.33E+03	1.77E+04	9.82E+05	1483.99	16.78
725	3.77E+01	2.01E+02	3.81E+03	3.02E+04	1.81E+06	1570.11	10.76

Table 2 (*Continued*)

Temperature (°C)	^{36}Ar	^{37}Ar	^{38}Ar	^{39}Ar	^{40}Ar	Age (Ma)	1 S.D.
750	4.25E+01	2.32E+02	4.41E+03	3.31E+04	1.95E+06	1551.43	4.41
775	1.77E+01	2.28E+02	4.28E+03	3.37E+04	2.00E+06	1560.33	2.23
800	2.34E+01	2.66E+02	3.93E+03	2.98E+04	1.73E+06	1534.06	2.14
850	1.63E+01	3.85E+02	4.04E+03	3.22E+04	1.87E+06	1541.56	2.13
900	2.09E+01	4.78E+02	5.27E+03	4.00E+04	2.30E+06	1528.62	1.38
950	2.55E+01	3.64E+02	4.64E+03	3.65E+04	2.13E+06	1542.39	2.09
975	2.06E+01	2.05E+02	2.59E+03	2.06E+04	1.19E+06	1533.33	4.62
1000	3.12E+01	1.35E+02	1.50E+03	1.23E+04	7.04E+05	1512.12	6.96
1050	4.16E+01	1.58E+02	2.06E+03	1.68E+04	9.71E+05	1526.46	5.41
1100	7.23E+01	2.97E+02	4.38E+03	3.45E+04	1.99E+06	1521.21	2.31
1150	1.19E+02	2.76E+02	8.61E+03	6.79E+04	3.93E+06	1528.51	4.88
1175	1.43E+02	1.68E+02	8.01E+03	6.31E+04	3.65E+06	1526.48	1.42
1200	1.47E+02	5.41E+01	3.23E+03	2.61E+04	1.55E+06	1533.73	7.87

Age (Ma): the age calculated using the following decay constants: $\lambda_\varepsilon = 0.581 \times 10^{-10}$ per year; $\lambda_\beta = 4.961 \times 10^{-10}$ per year; ^{40}K/K = 0.01167 atom% (Steiger and Jäger, 1977).

drift configuration (D'Agrella-Filho et al., 1996; D'Agrella-Filho and Pacca, 1998). Similar rocks from the Jequié block, located right to the west of the sampling area (D'Agrella-Filho and Pacca, 1994), show northeasterly (westerly) declination, downward (upward) inclination, directions. The age of this component is constrained by ~2000 Ma ^{40}Ar/^{39}Ar plateau ages on hornblendes and biotites separated from the same rocks (Sletten, 1989), thus attesting to the older nature of the characteristic magnetization found in the granulitic rocks of Salvador.

Sites SA8b and SA16b give a positive contact test confirming the primary, thermoremanent nature of the characteristic magnetizations obtained in the studied unmetamorphosed dikes. The results obtained in the site SA17 are incomplete since the host-rock far from the dike shows a very unstable magnetic behavior, probably due to the low-coercivity of remanence carriers in these rocks (Figs. 5c and 6). However, the similarity between the directions found in the dike and in the host-rock samples located nearby can be likely attributed to the thermal influence of the intrusion. Three lines of evidence reinforce this interpretation: (i) the unblocking temperatures above 500 °C (Fig. 4a) and the high temperature exsolutions of the remanence carriers in the dike; (ii) the progressive change in hysteresis parameters of granulites towards the dike, suggesting mineralogical transformation induced by the intrusion (Fig. 6); and (iii) the ^{40}Ar/^{39}Ar step heating analysis at the same site which will be presented in the following section.

6. ^{40}Ar/^{39}Ar analysis

The ^{40}Ar/^{39}Ar analyses (Table 2) were performed in the Department of Geological and Geophysical Sciences of Princeton University (USA) in 1989. Concentrates of slightly sericitized plagioclase from sample JA51, and biotite from samples JA57, JA58 and JA60 were irradiated at the McMaster University reactor in Hamilton, Ontario (Canada) using the Mmhb-1 standard (520.4 Ma; Samson and Alexander, 1987).

The experimental procedures at the argon lab are described by Onstott and Peacock (1987). The samples were crushed with a mortar and pestle and sieved to obtain a favorable grain size (usually between 80 and 100 sieve size). Plagioclase separates were extracted using a Frantz Isodynamic magnetic separator, followed by ultrasonic cleaning (10 min), and final concentration with heavy liquids. Additional treatment with 0.1 N hydrofluoric acid was used to reduce any influence of weathering from the plagioclase surface. Biotite was separated using the technique of "paper shaking", followed by hand picking.

Plagioclase (sample JA51) was laser step-heated following the procedure of Cumbest et al. (1994). The biotite samples (JA57, JA58 and JA60) were transferred to quartz boats and were step-heated in a furnace, using an on-line extraction-computer system. The isotopic ratios were measured in a UG-WAT GD150 mass spectrometer, and the errors of the ^{40}Ar/^{39}Ar dates are at ±2 standard deviation level. The ^{40}Ar/^{39}Ar ages were calculated following the

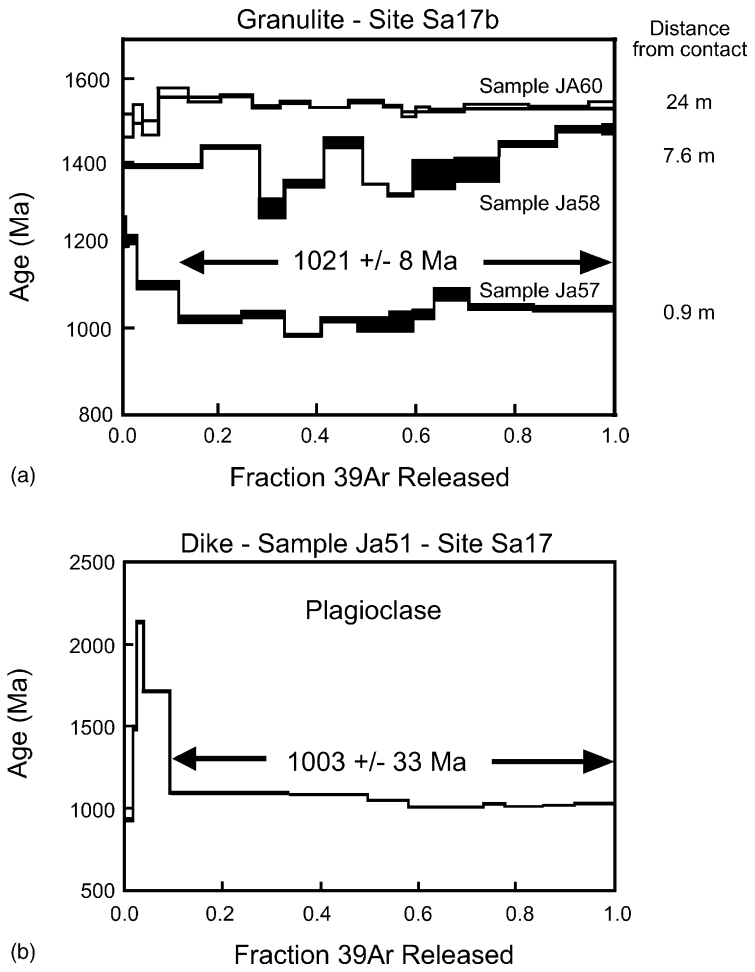


Fig. 12. Apparent ages as a function of cumulative ^{39}Ar released. (a) biotite separates of samples from country rocks (granulites) situated at different distances from the contact (0.90, 7.6 and 24 m) with a ~30 m thick dike, (b) plagioclase separate from the dike JA51. Apparent ages are corrected for non-radiogenic Ar having atmospheric isotopic composition ($^{40}Ar/^{36}Ar = 295.5$). Error bars are stated at 2 standard deviation-level and do not reflect uncertainty in the *J*-value, age of standard and errors in the interference corrections.

procedures of Wood et al. (1991), and integrated ages were calculated according to Onstott and Peacock (1987).

$^{40}Ar/^{39}Ar$ analysis on biotite from granulite JA51 (at ~90 cm from the dike's contact in site SA17) yields a $^{40}Ar/^{39}Ar$ integrated age of 1021 ± 8 Ma (Fig. 12a). Plagioclase from this dike (Table 2) yields an integrated age of 1003 ± 33 Ma (Fig. 12b) which agrees within error with the biotite age. In both cases the initial $^{40}Ar/^{36}Ar$ values (not shown in the figure) are higher than the atmospheric value indicating excess of radiogenic ^{40}Ar, as mirrored from the anomalously high peaks of the first steps of the analyses (Fig. 12).

Two other samples from granulites collected at 7.6 (JA58) and 24 m (JA60) from the dike's contact, yield progressively older apparent ages, but only sample JA60 defines a pseudo-plateau age of ca. 1550 Ma (Fig. 12a). Therefore, these apparent ages may reflect partial Ar loss of the mineral systems, induced by the dike emplacement. For instance, several K–Ar ages obtained for the country rocks far from any mafic dike are much older, ranging between 1800 and 1700 Ma (Teixeira and Canzian, 1994).

7. Discussion

Paleomagnetic results are summarized in Tables 1 and 3. The corresponding paleomagnetic poles are referred to the present-day position of the São Francisco craton. They were corrected from a 2° counterclockwise rotation about a rotation pole at 36°04′W, 8°11′S to account for the development of the Reconcavo-Tucano-Jatobá rift (Fig. 2a) at Cretaceous times (Milani and Davison, 1988).

7.1. Paleomagnetic poles of the Salvador unmetamorphosed dikes

The first steps of the $^{40}Ar/^{39}Ar$ analyses of plagioclase JA51 and biotite JA57 (Fig. 12) are clearly influenced by excess radiogenic ^{40}Ar (e.g., Lee et al., 1990). However, most of the $^{40}Ar/^{39}Ar$ ratios resulted in comparable ages (1021 ± 8 Ma and 1003 ± 33 Ma), which are interpreted to be the emplacement age of the unmetamorphosed dikes in Salvador. The range of

Table 3
Mean directions and poles obtained for the Salvador rocks

Name	N	Site mean directions				Paleomagnetic pole				Rotated pole		
		Dec (°)	Inc (°)	K	α_{95} (°)	P_{long} (°E)	P_{lat} (°N)	κ	A_{95} (°)	P_{long} (°E)	P_{lat} (°N)	Age (Ma)
S_{AR}	7	290.7	64.7	65.2	7.5	282.6	5.6	44.3	9.2	283.0	7.0	
S_{AN}	8	110.8	−79.3	44.4	8.4	122.7	6.4	14.8	14.9	122.8	5.7	1021 ± 8, 1003 ± 33
S	15	110.8	−72.5	38.0	6.3	113.1	0.7	15.4	10.1	113.4	−0.4	
S_B	5	104.6	−36.0	35.6	13.0	74.9	−8.3	40.1	12.2	75.3	−10.2	
S_C	3	168.6	65.6	67.3	15.1	334.9	−52.8	31.8	22.2	332.6	−53.1	
S_{MN}	9	294.9	−42.5	21.6	11.3	209.0	28.2	20.3	11.7	208.8	30.0	Paleoprot[a]
S_{MR}	4	26.8	60.2	27.8	17.7	344.7	30.4	12.9	26.7	346.1	29.7	Paleoprot[a]

Name: pole denomination: unmetamorphosed dikes—S_{AR} (A component with steep positive inclination), S_{AN} (A component with steep negative inclination), S (A component, both polarities), S_B (B component), S_C (C component); metamorphic samples—S_{MN} (northwest direction with negative inclination, Fig. 7b), S_{MR} (northeast direction with positive inclination, Fig. 7b), N, number of sites; Dec, declination; Inc, inclination; K, α_{95}, κ, A_{95}, Fisher's statistic parameters: P_{long}, pole longitude; P_{lat}, pole latitude; rotated pole, pole position after 2° clockwise rotation about the rotation pole at 36°04′W, 8°11′S (Milani and Davison, 1988).

[a] Supposedly Paleoproterozoic.

Table 4
Mean directions and paleomagnetic poles (1.1–1.0 Ga) for the São Francisco craton

Name	N	Site mean directions				Paleomagnetic poles					Reference
		Dec (°)	Inc (°)	K	α_{95} (°)	P_{long} (°E)	P_{lat} (°N)	κ	A_{95} (°)	Age (Ma)	
I_N	17	60.0	−66.8	190	3	100.4	30.3	79	4	1012 ± 24	D'Agrella-Filho et al. (1990)
O_N	31	82.4	−71.0	27	5	107.0	16.1	12	8		D'Agrella-Filho et al. (1990)
IC	23	99.0	−71.9	27	6	111.0	7.7	11	10		D'Agrella-Filho et al. (1990)
S_{AN}	8	110.8	−79.3	44	8	122.8	5.7	15	15	1021 ± 8, 1003 ± 33	This work
S_{AR}	7	290.7	64.7	65	8	283.0	7.0	44	9		This work
O_R	18	298.8	60.7	30	6	280.2	9.5	17	9	1078 ± 18	D'Agrella-Filho et al. (1990)
I_N	17					100.4	30.3	79	4	1012 ± 24	D'Agrella-Filho et al. (1990)
S_{AN}–IC/O_N	62					110.6	11.7	11	6	1021 ± 8	This work
S_{AR}–O_R	25					281.0	8.8	21	7	1078 ± 18	This work

Name: pole denomination: unmetamorphosed dikes—I_N (Ilhéus dikes), O_N, O_R (Olivença dikes), IC (Itaju do Colônia dikes), S_{AN}, S_{AR} (Salvador dikes), S_{AN}–IC/O_N (combined pole—mean of VGPs including normal polarity Olivença dikes, normal polarity Salvador dikes, and both normal and reversed polarity Itaju do Colônia dikes), S_{AR}–O_R (combined mean—mean of VGPs including reversed polarity Salvador dikes and reversed polarity Olivença dikes); N, number of sites; Dec, declination; Inc, inclination; K, α_{95}, κ, A_{95}, Fisher's statistical parameters; P_{long}, pole longitude; P_{lat}, pole latitude.

these ages agrees within error with the 1080–1010 Ma ages reported for the E–W dike swarm of Ilhéus and Olivença (Renne et al., 1990). The $^{40}Ar/^{39}Ar$ data suggest that this swarm of dikes have not experienced any significant thermal overprint that could have reset their paleomagnetic signature since their emplacement.

The mean directions for component A (normal and reverse) are not symmetrically opposed (Fig. 7). The statistical test proposed by McFadden and Lowes (1981) for discrimination of mean directions from Fisher (1953) distributions shows that these mean directions are statistically different at the 95% probability level. Hence, the normal and reverse A components give two paleomagnetic poles, S_{AN} (122.7°E, 6.4°N, $K = 14.8$, $N = 8$ and $A_{95} = 14.9°$) and S_{AR} (282.6°E, 5.6°N, $K = 44.3$, $N = 7$ and $A_{95} = 9.2°$), respectively. These poles (Table 4) match the APW path defined for the São Francisco/Congo unit for the interval 1080–1010 Ma (Fig. 13). After inclusion of the Salvador data in this path, a low drift-rate between 1080 and 1020 Ma for the São Francisco/Congo unit is defined based on the $^{40}Ar/^{39}Ar$ integrated age of 1020 Ma obtained for dike SA17 (pole S_N). This regime was followed by a high drift-rate between 1020 and 1010 Ma. Accordingly, this trajectory is marked by reversals (Fig. 13).

Component B was revealed in five dikes from the Valéria quarry, while component A_N appeared in seven dikes (and in granulitic rocks from site SA17b). They have similar declinations, but their mean inclinations differ by about 40° (Fig. 7a and Table 3). Either these dikes have different ages and the difference in inclination represents latitudinal plate movements, or they have similar ages and the difference in inclination represents relative tectonic movement of the sampled areas. Since relative tectonic movement of the sampled areas during the Proterozoic is not supported by geological evidence, and radiometric ages are not available for the dikes in the Valeria quarry (see Fig. 2b), an interpretation about component B is still open.

Only three dikes yielded component C, one of which is from the Ondina area (Fig. 7a). This dike was emplaced at 920 Ma according to its U–Pb age on badde-

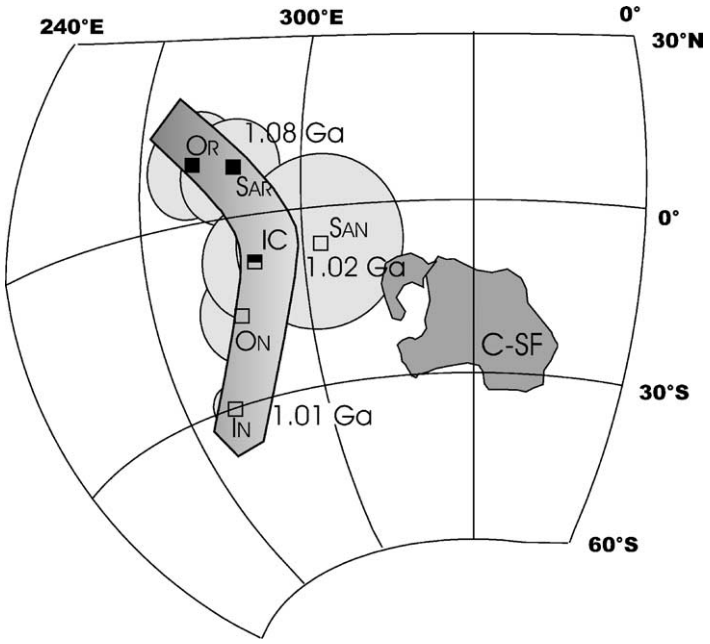


Fig. 13. Apparent polar wander path for Congo/São Francisco (C-SF) craton defined by poles from Ilhéus (I_N), Olivença (O_N, O_R), Itaju do Colônia (IC) and Salvador (S_{AR}, S_{AN}) mafic dike swarms. Empty (full) symbols represent poles with "normal" ("reverse") polarities. Semi-full symbols represent poles with both polarities. Congo craton rotated to the São Francisco craton, according to the Euler rotation parameters: 45.5°N, −32.2°E, −58.2°. Reconstruction done with the GMAP package of T.H. Torsvik and M. Smethurst.

leyite (Heaman, 1991). Despite this well constrained age of emplacement, a larger number of sites yielding this component is still necessary to average out secular variation and to allow comparison with other published poles of similar age.

7.2. Distinctness of group mean VGPs

D'Agrella-Filho et al. (1990) have applied the statistical test of McFadden and Lowes (1981) to the Ilhéus (I_N), Olivença (O_N, O_R) and Itaju do Colônia (IC) mean directions. They have shown that the Ilhéus (I_N) and Olivença (O_R) mean directions are statistically different from each other and from the Olivença (O_N) and Itaju do Colônia (IC) mean directions at the 95% probability level. However, the Olivença (O_N) and Itaju do Colônia (IC) mean directions are statistically indistinguishable at the 95% probability level, and its mean pole (named IC/O_N pole) lies at 108.7°E, 12.6°N ($N = 54$, $A_{95} = 6.0°$, $K = 11.2$).

Since the Salvador and Ilhéus-Olivença-Itaju do Colônia areas are separated by more than 400 km this test cannot be applied to their mean directions. Therefore, the test was applied to their VGPs (virtual geomagnetic poles) supposing they obey a Fisherian distribution. Firstly, the test shows that the Salvador S_{AN} and S_{AR} poles are statistically different at the 95% probability level (Table 5). Comparison of these individual poles (S_{AR} and S_{AN}) with those from the Olivença (O_R) and the combined Olivença and Itaju do Colônia (IC/O_N) (see above) shows that the poles S_{AR} and O_R as well as poles S_{AN} and IC/O_N poles are statistically indistinguishable at the 95% probability level (Table 5).

Combination of these paleomagnetic data (Table 4) results in three well dated paleomagnetic poles for Congo/São Francisco: pole I_N (100.4°E, 30.3°N, $N = 17$, $A_{95} = 4.0°$) with an age of 1012 ± 24 Ma, pole S_{AN}–IC/O_N (110.6°E, 11.7°N, $N = 62$, $A_{95} = 5.6°$) with an age of 1021 ± 8 Ma and pole S_{AR}–O_R (281.0°E, 8.8°N, $N = 25$, $A_{95} = 6.5°$) with an age of 1078 ± 18 Ma. These paleomagnetic poles satisfy six out of the seven criteria of quality defined by Van der Voo (1990). Therefore, they are classified as key paleomagnetic poles, and can be used to constrain the paleogeography of this block at the end of Mesoproterozoic.

7.3. Tectonic implications for the Rodinia Supercontinent

Fig. 14 shows paleogeographic reconstructions for the Congo/São Francisco unit, using the three key paleomagnetic poles calculated for the São Francisco craton (I_N, S_{AN}–IC/O_N, S_{AR}–O_R). Kalahari and Laurentia are positioned according to selected paleomagnetic poles (Table 6). Cratonic units surrounding Laurentia at their western, northern and eastern margins are as in Dalziel et al. (2000), although alternative reconstructions are also possible (e.g., Karsltrom et al., 1999; Wingate et al., 2002; Pisarevsky et al., 2003; Meert and Torsvik, 2003). In fact, the position of almost all the peri-Laurentia fragments is critical. At least three configurations for the Australia-East Antarctica-Laurentia connection have been proposed, named through the acronyms SWEAT, AUSWUS, and AUSMEX (e.g., Wingate et al., 2002). Although geological and geochronological evidence favor a possible connection between Amazonia, Baltica and Laurentia during Mesoproterozoic times, different geodynamic interactions between these units have been proposed (e.g., Sadowski and Bettencourt, 1996; Weil et al.,

Table 5
Comparison of mean VGPs

Pole 1	Pole 2	R_1	R_2	R	N	U	$(1/P)^{1/(N-2)} - 1$ ($P = 0.05$)	$(1 - P) \times 100$ (%)
S_{AR}	S_{AN}	6.8649	7.5271	14.0924	15	0.4877	0.2592	99
S_{AR}	O_R	6.8649	16.9973	23.8514	25	0.0095	0.1391	20
S_{AN}	IC/O_N	7.5271	49.283	56.5704	62	0.0461	0.0512	93

Length of the vector sum: R_1 (pole 1); R_2 (pole 2); R (resultant of poles 1 and 2); N: total of sites in the two populations; $U = [R_1 + R_2 - R^2/(R_1 + R_2)]/2(N - R_1 - R_2)$. The hypothesis that two populations of VGPs (N_1 and N_2), with similar precisions (k), share a common true mean pole may be rejected at the level of confidence P, if $U > (1/P)^{1/(N-2)} - 1$ ($P = 0.05$). It is supposed that each population obey the Fisher (1953) distribution (McFadden and Lowes, 1981). The last column presents the probability of distinctness $[(1/P)^{1/(N-2)} - 1 = U]$.

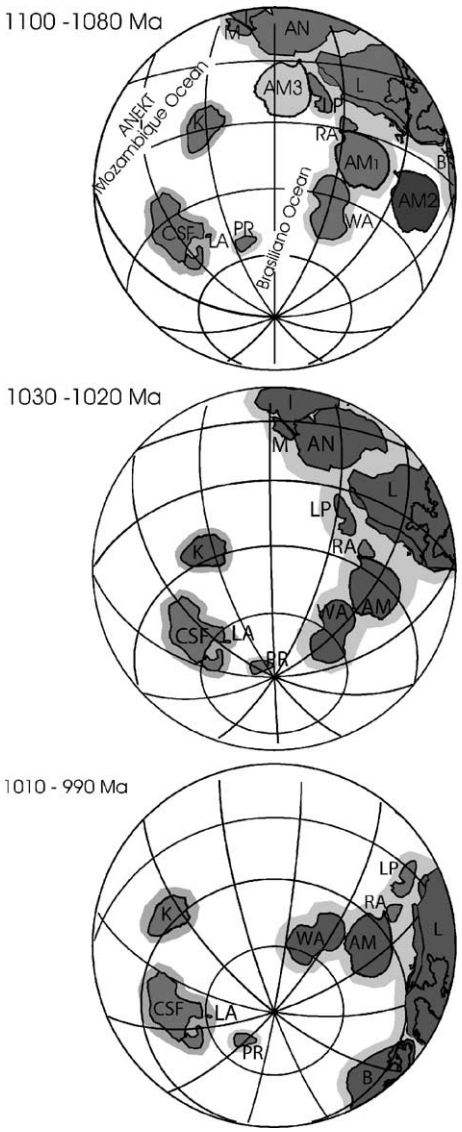


Fig. 14. Paleogeographical reconstructions using poles from Table 6. East Gondwana and Baltica as in Dalziel et al. (2000). Amazonia (AM_1), West Africa and Rio de la Plata blocks as in Weil et al. (1998). Other two paleogeographies of Amazonia, according to Dalziel et al. (2000) (AM_2) and Tohver et al. (2002) (AM_3) reconstructions are also shown in the 1100–1080 Ma reconstruction. Abbreviations: L, Laurentia; M, Madagascar; I, India; AN, East Antarctica; A, Australia; K, Kalahari; CSF, Congo/São Francisco; LA, Luis Alves; PR, Paraná; RA, Rio Apa; LP, Rio de la Plata; B, Baltica; WA, West Africa. The main feature of these reconstructions is the large Brasiliano Ocean between São Francisco/Congo and Amazonia/West Africa cratons during that time. Reconstructions done with the GMAP package of T.H. Torsvik and M. Smethurst.

1998; Buchan et al., 2000; Dalziel et al., 2000; Elming and Mattsson, 2001; Tohver et al., 2002; Pisarevsky et al., 2003; Meert and Torsvik, 2003). Fig. 14 shows three possible configurations for Amazonia, according to Dalziel (1997), Weil et al. (1998) and Tohver et al. (2002). Connections along Greenland (Dalziel, 1997) and Labrador (Weil et al., 1998) were initially proposed based on the fit of geological provinces and scarce virtual geomagnetic poles. Recently, Tohver et al. (2002) suggested a link between Amazonia and Laurentia along the Llanos Orogeny based on the 1.2 Ga Nova Floresta pole (Fig. 14). However, we prefer the Weil et al.'s (1998) fit for two reasons. First, the Tohver et al.'s reconstruction was based on a single Amazonian paleomagnetic pole and thus lacks a longitudinal control. Moreover, unpublished 1.2–1.1 Ga paleomagnetic poles recently obtained for Amazonia (D'Agrella-Filho et al., 2003) are coherent with an oblique collision of this craton with Laurentia along the Labrador region, which culminated by about 1.1 Ga in a paleogeography similar to that proposed by Weil et al. (1998). No Mesoproterozoic paleomagnetic poles are actually available for the Rio de La Plata craton and its position close to Laurentia (Dalziel, 1997), is based on contested geological grounds (see Hoffman, 1999). Similarly, the position of India is yet unconstrained (e.g., Dalziel et al., 2000; Torsvik et al., 2001a,b). The position of Kalahari in the Rodinia reconstructions is also controversial. Based on 1100–1000 Ma APW paths for Kalahari and Laurentia, Weil et al. (1998) and D'Agrella-Filho et al. (1998) placed Kalahari close to southern Laurentia (Fig. 1). Dalziel et al. (2000) present geochronological and geological evidence that reinforce a possible connection between Kalahari and Laurentia along the grenvillian Namaqua/Natal (Kalahari) and Lhanos (Laurentia) orogenic belts. However, this is contested by Powell et al. (2001) who argued that presently available paleomagnetic data imply a latitudinal separation of $\sim 30^{\circ}$ between Kalahari and Laurentia at about 1100 Ma. Alternatively, they suggest that Kalahari lies close to East Antarctica; the Namaqua-Natal belt facing outboard and away from Laurentia. In our reconstruction (Fig. 14) we preferred to place Kalahari close to Congo/São Francisco craton, based on a possible connection of these cratons at 820 Ma along the Zambezi belt (Hanson et al., 1998; Vinyu et al., 1999).

Table 6
Selected paleomagnetic poles used in reconstructions of Fig. 14

	Age (Ma)	Pole		α_{95} (°)	Quality factor	Reference
		°N	°E			
Laurentia						
Mean Logan dikes	1100	35	181	10	6	Weil et al. (1998)
Lower Normal, Upper Osler Group	1095–1101	34	178	9	4	Weil et al. (1998)
Portage Lake volcanics	1094–1098	27	181	2	4	Weil et al. (1998)
Mamainse Point volcanics	1083–1097	38	188	1	4	Weil et al. (1998)
Chipman Lake carbonatites	1090	38	186	8	5	Weil et al. (1998)
Mamainse Point Intrusive Unit	1083–1088	24	166	31	3	Weil et al. (1998)
Clay-Howells carbonatite Complex	1060–1090	27	179	7	5	Weil et al. (1998)
Michipicoten Island volcanics	1088	25	175	8	4	Weil et al. (1998)
Lake Shore Traps	1085–1089	22	181	5	5	Diehl and Haig (1994) and Davis and Paces (1990)
Upper North Shore volcanics	1095–1099	32	184	5	5	Halls and Pesonen (1982) and Davis and Green (1997)
Mean pole	1100–1080	30.3	179.7	4.8		
Eileen sandstone	1020	20	156	10	3	Weil et al. (1998)
Middle River sandstone/shale	1020	25	148	9	3	Weil et al. (1998)
Mean pole	1020	22.5	152.1	19.6		
Chequamegon sandstone	1010	−12	178	5	4	McCabe and Van der Voo (1983)
Jacobsville sandstone	1010	−9	183	6	5	Weil et al. (1998)
Grenville thermochron zone A	1000	1	159	6	3	Weil et al. (1998)
Mean pole	1000–1010	−6.8	173.3	22		
Congo/São Francisco						
Olivença—O_R/Salvador—S_{AR}	1080	−9	101	7	6	This work
Olivença—O_N/Salvador—S_{AN}/Itaju do Colônia/IC	1020	12	111	6	6	This work
Ilhéus—I_N	1010	30	100	4	6	This work
Amazonia						
Nova Floresta gabbros and basalts	1194–1203	24.6	164.6	5.5	5	Tohver et al. (2002)
Kalahari						
Unkondo igneous province	1108–1091	−65	223	6	3	Powell et al. (2001)
O'Okiep intrusives, S. Africa	1030	−15	155	15	1	Weil et al. (1998)
Central Namaqua metamorphic zone	980–1020	−8	150	10	3	Onstott et al. (1986) and Renne et al. (1990)

The Congo/São Francisco plate has been placed to the south–southeast of the present-day margin of Laurentia in previous Rodinia reconstructions (Dalziel et al., 2000; Weil et al., 1998; D'Agrella-Filho et al., 1998) (Fig. 1). However, recent geological and geochronological data from South America and Africa suggest that a large ocean was formed between this unit and the Amazonia/West Africa craton in the Meso-Neoproterozoic (e.g., Pimentel et al., 1999). This interpretation is supported by juvenile granitoid magmatism recorded all along the present-day central Brazil and Western Africa which has started at 940 (Pimentel et al., 1999) and 830 Ma (Caby, 1989), respectively. From this evidence we placed the Congo/São Francisco plate far from the main Rodinia Supercontinent in Fig. 14. Also shown in Fig. 14 is the Paraná block which is the basement of the Phanerozoic Paraná basin. Subduction-related granitic magmatism recognized in the northern border of the Paraná block suggests ocean closure between this block and the Congo/São Francisco craton at about 670 Ma (Campos-Neto and Caby, 1999).

8. Conclusions

Paleomagnetic analysis on late Mesoproterozoic mafic dikes from the Salvador area (São Francisco craton) yield paleomagnetic directions that combined with those from Ilhéus, Olivença and Itaju do Colônia area (D'Agrella-Filho et al., 1990) allow us to calculate three well dated poles: pole I_N (100.4°E, 30.3°N, $N = 17$, $A_{95} = 4.0°$) with an age of 1012 ± 24 Ma, pole S_{AN}–IC/O_N (110.6°E, 11.7°N, $N = 62$, $A_{95} = 5.6°$) with an age of 1021 ± 8 Ma and pole S_{AR}–O_R (281.0°E, 8.8°N, $N = 25$, $A_{95} = 6.5°$) with an age of 1078 ± 18 Ma. They are considered to represent primary thermo-remanent magnetizations, and can be ranked as high-quality ($Q = 6$; Van der Voo, 1990) paleomagnetic poles. These poles are used to constrain the paleogeography of the Congo/São Francisco craton (Fig. 14). The main feature of paleogeographic reconstructions for the Meso-Neoproterozoic time is that the Congo/São Francisco plate and other South American and African continental fragments did not take part in Rodinia. At that time this landmass was separated from the other continental fragments by a large ocean (named Brasiliano Ocean), which closed between 940 and 630 Ma, during the assembly of Gondwana.

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